

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. (CL) Examination

November 2013

CL 214 Fluid Mechanics II

Date: 14.11.2013, Thursday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

- Q - 1 (a) Derive equation for flow of viscous fluid between two plates when one plate is moving and other at rest. [07]
- Q - 2 (a) Explain terms (1) Rapidly varied flow (2) Gradually varied flow [04]
- (b) Find the rate of flow of water through a V-shaped channel having height of 4.0 m and v-shape angle of 60° . Take value of $C=55$ and slope of the bed 1 in 1000. [04]
- (c) What is specific energy curve? Draw specific energy curve and then derive expressions for critical depth and critical velocity. [06]

OR

- Q - 2 (a) Define the term most economical section of a channel. What are the conditions for the rectangular channel of the best section? [02]
- (b) A trapezoidal channel has side slopes of 3 horizontal to 4 vertical and slope of its bed is 1 in 2000. Determine the optimum dimensions of the channel, if it is to carry water at $0.5 \text{ m}^3/\text{s}$. Take chezy's constant as 80. [04]
- (c) Find the discharge through a circular pipe of diameter 3.0 m, if the depth of water in the pipe is 1.0 and the pipe is laid at a slope of 1 in 1000. Take the value of Chezy's constant as 70. [08]
- Q - 3 (a) Explain propagation of pressure waves with neat sketch. [07]
- (b) Calculate the stagnation pressure, temperature and density at the stagnation point on the nose of a plane, which is flying at 800 Km/hour through still air having a pressure 8.0 N/cm^2 (abs.) and temperature -10° C . Take $R=287 \text{ J/KgK}$ and $K=1.4$. [07]

OR

- Q - 3 (a) Enlist the basic equations of compressible flow and derive expression for bernoulli's equation when process is isothermal. [05]
- (b) Find the velocity of air flowing at the outlet of a nozzle, fitted to a large vessel which [07]

contains air at a pressure of 294.3 N/cm^2 (abs.). Take $k=1.4$ and $K=287 \text{ J/kgK}$

- (c) Find the speed of the sound wave in air at sea-level where the pressure and temperature are 10.1043 N/cm^2 (abs.) and 15°C respectively. Take $R=287 \text{ J/kgK}$ and $K=1.4$. [02]

SECTION – II

- Q - 4 (a) Explain separation of boundary layer with neat sketch and methods of preventing the same. [07]

- Q - 5 (a) Derive Darcy-Weishbach equation. [06]

- (b) Calculate (1) the pressure gradient along flow, (2) the average velocity, and (3) the discharge for an oil of viscosity 0.02 Ns/m^2 flowing between two stationary parallel plates 1 m wide maintained 10 mm apart. The velocity midway between the plates is 2 m/s. [06]

- (c) Define boundary layer thickness and displacement thickness. [02]

OR

- Q - 5 (a) Derive von karmann integral equation for boundary layer flow. [07]

- (b) Determine (1) the pressure gradient, (2) the shear stress at the two horizontal parallel plates and (3) the discharge per metre width for the laminar flow of oil with a maximum velocity of 2 m/s between two horizontal parallel fixed plates which are 100 mm apart. $M=2.4525 \text{ Ns/m}^2$. [07]

- Q - 6 (a) Classify and explain different types of models. [04]

- (b) Define following terms: [03]

1. Reynold's Number.

2. Froude's Number.

- (c) The efficiency η of a fan depends on density ρ , dynamic viscosity μ of the fluid, angular velocity ω , diameter D of the rotor and the discharge Q . Express η in terms of dimensionless parameters. [07]

OR

- Q - 6 (a) State Buckingham's π -theorem. [03]

- (b) Differentiate between model and prototype. [03]

- (c) The pressure difference Δp in a pipe of diameter D and length l due to viscous flow depends on the velocity V , viscosity μ and density ρ . Using Buckingham's π -theorem, obtain an expression of Δp . [08]